IMPERIAL

Predicting tropical cyclone landfall timing and intensity decay

using **TropiCycloneNet Dataset** (TCND) and **U-FNO**

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ELEC70127: ML for Tackling Climate Change

GitHub Link

Agenda

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3 Model Architecture

4 Feature Ablation

5 Model Results

6 Cross-Basin Generalization

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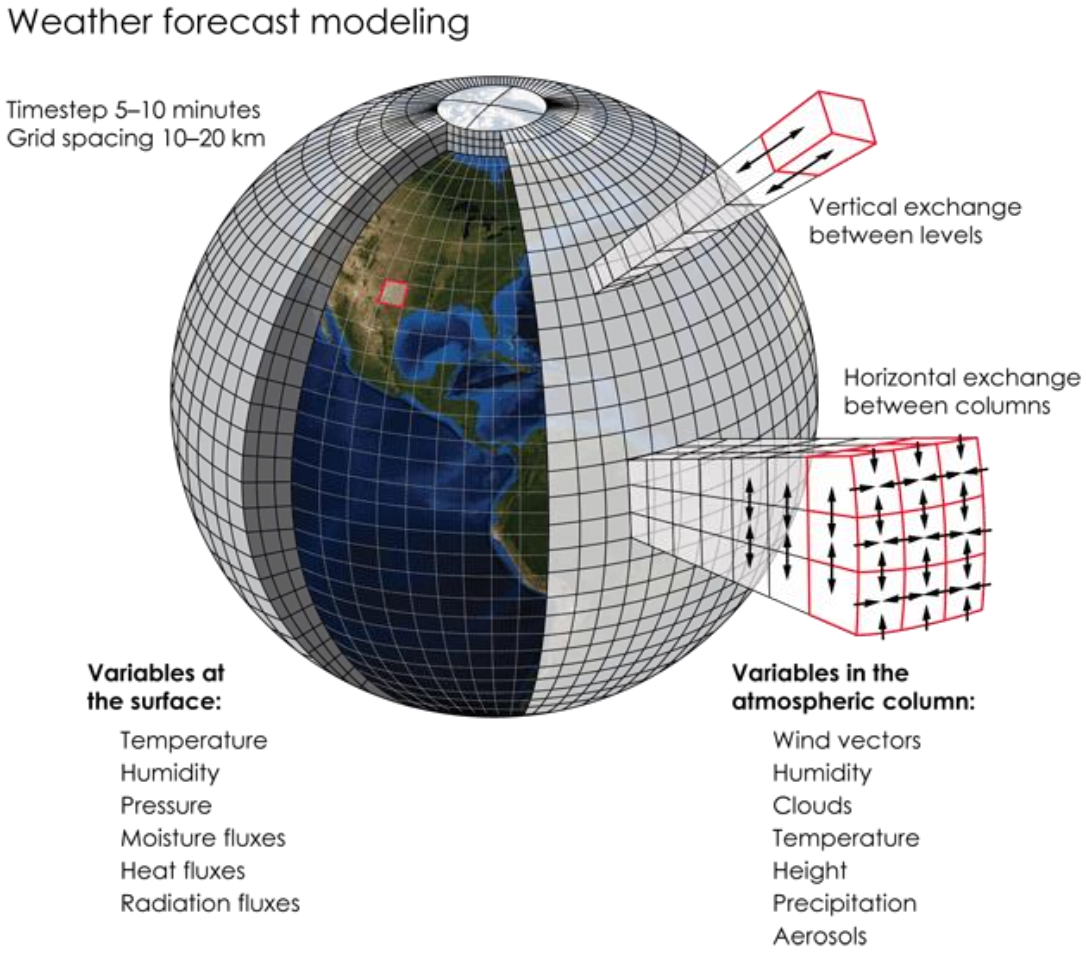
Introduction

Motivating Problem

There is a prediction gap between slow computation and an urgent need for data-driven, rapid early warnings.

The gold standard for cyclone prediction relies on numerical Weather prediction (NWP):

- Used in models like the **GFS** (Global Forecast System) or the **ECMWF** (European Centre for Medium-Range Weather Forecasts).
- They integrate complex fluid dynamic and thermodynamic equations like the Navier-Stokes equations) across a 3d grid.
- Require hours to make a 7-day forecast.
- These models are perfect for regular weather forecasting, but are **too slow for making decisions under atmospheric hazard weather conditions.**[1]



1. Shi et al. (2025)

Image: https://www.climate.gov/maps-data/climate-data-primer/predicting-climate/climate-models

Rising sea surface temperatures are driving a surge in catastrophic Category 4 and 5 tropical cyclones, leading to an increasing necessity for tools that can adequately predict aspects of these storms. For the safety of the people,

NWPs are too SLOW.

The TropiCycloneNet Dataset (TCND) contains **years of data** and allows for a **modular integration of tabular, spatial, and environmental features.**

How can we use the TCND to predict the landfall timing and intensity decay of tropical storms, therefore improving coastal early-warning systems?

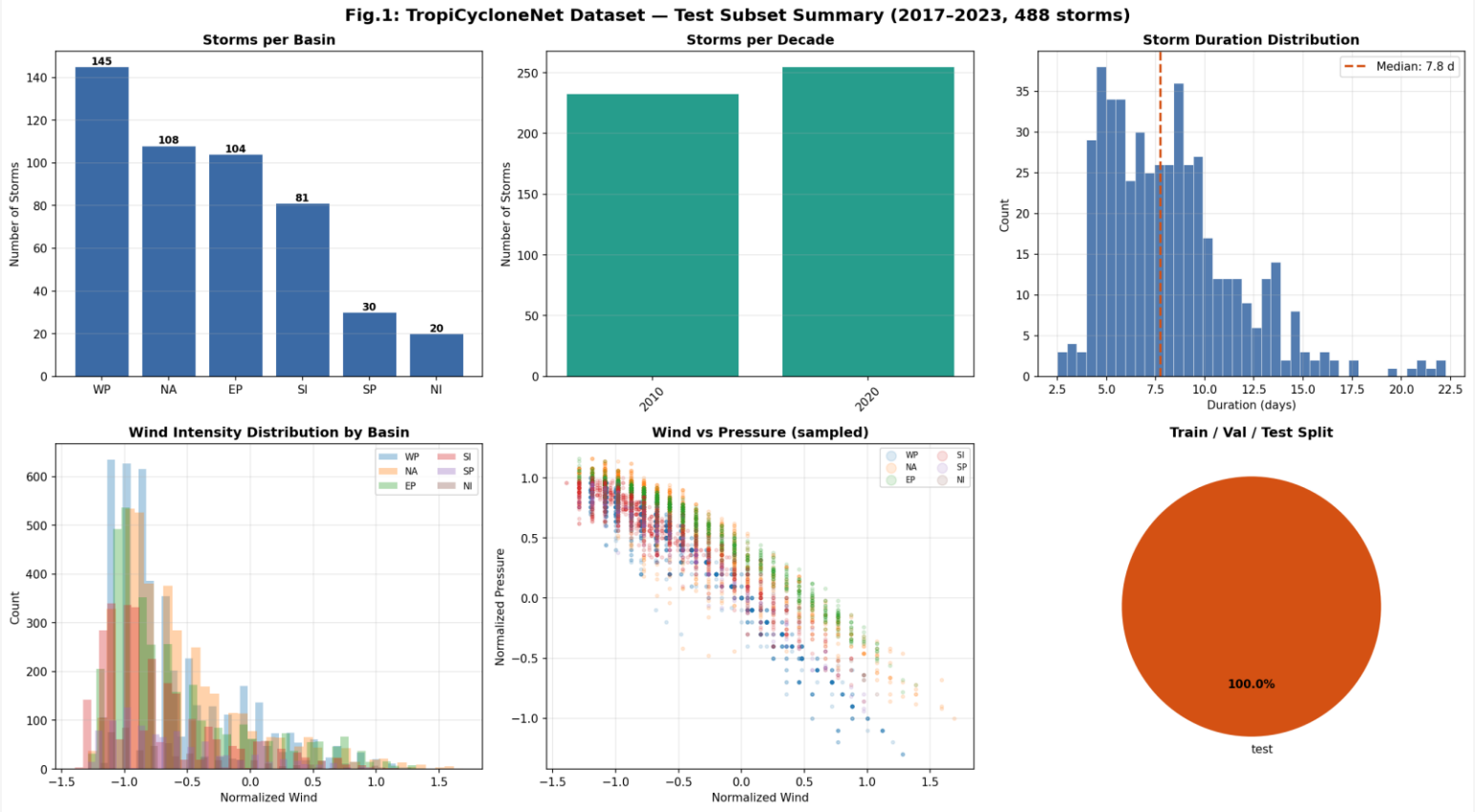
Exploratory Data Analysis

Understanding TCND*

*All analysis was done on the test dataset due to full dataset access issues.

Analysis
Comprehensive overview

- The TropiCycloneNet (TCND) test subset covers **488 storms (2017-2023).**
- WP/NA basins provide the highest density, but the inclusion of SI/SP basins ensures **the model isn't overfitted to a single hemisphere.**
- Storm durations vary significantly, with a **median of 7.8 days**.
- The "Wind vs. Pressure" scatter plot shows a **tight negative correlation.**
- When we build the model, we have to ensure it **learns physically grounded relationships** (higher wind = lower pressure) rather than statistical noise.



Analysis
Landfall overview

Post-Landfall Weakening:

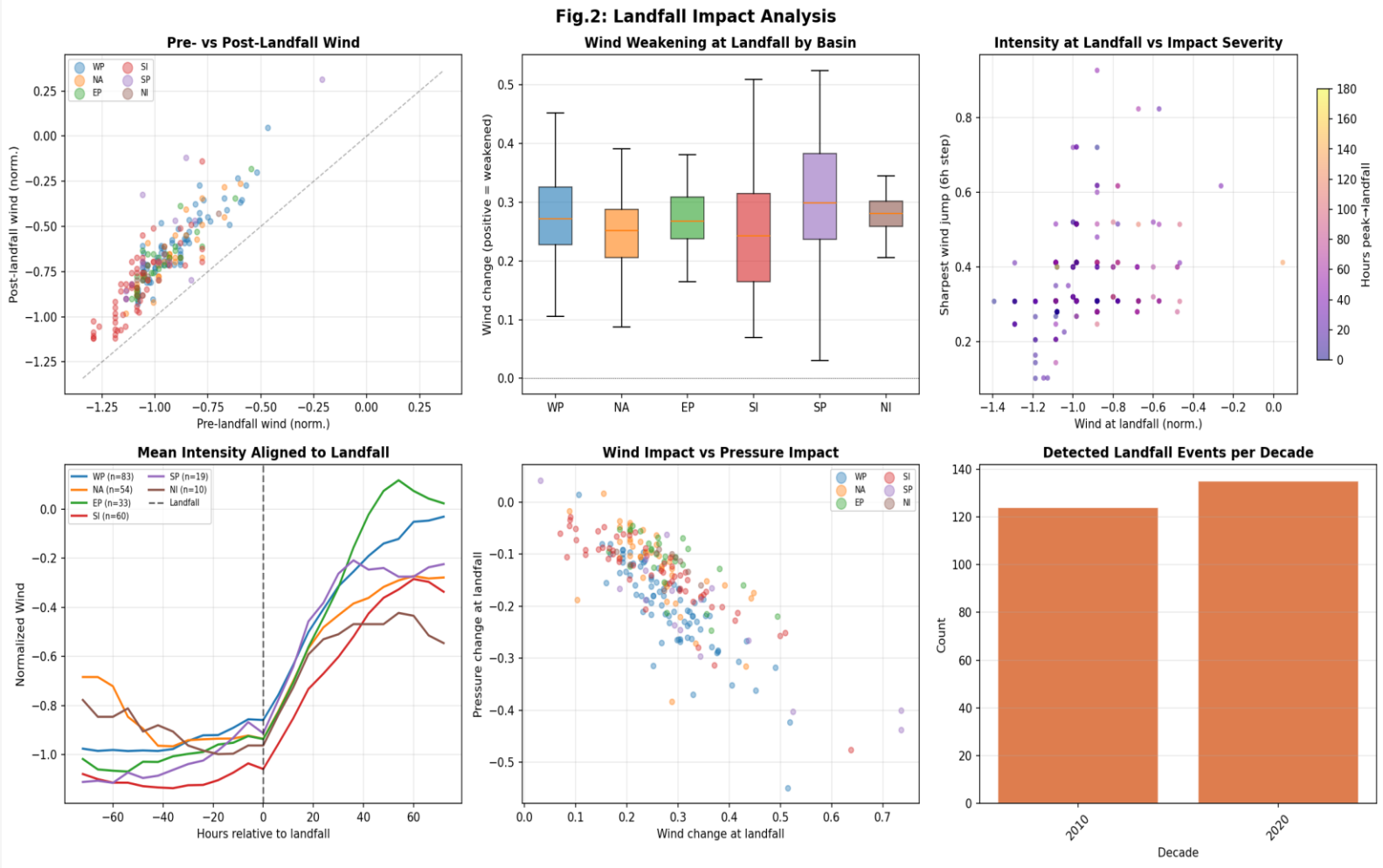
The "Mean Intensity Aligned to Landfall" plot shows a sharp, universal upward slope immediately after T=0.

Basin-Specific Nuances:

In the same plot, we observe that trajectories for each basin all have different starting points.

Wind vs. Pressure:

There is a clear, negative correlation that the model can learn. Given this, it motivates the need for a model that can learn patterns grounded in physics.



Analysis

Intensity decay overview

Basin Variance:

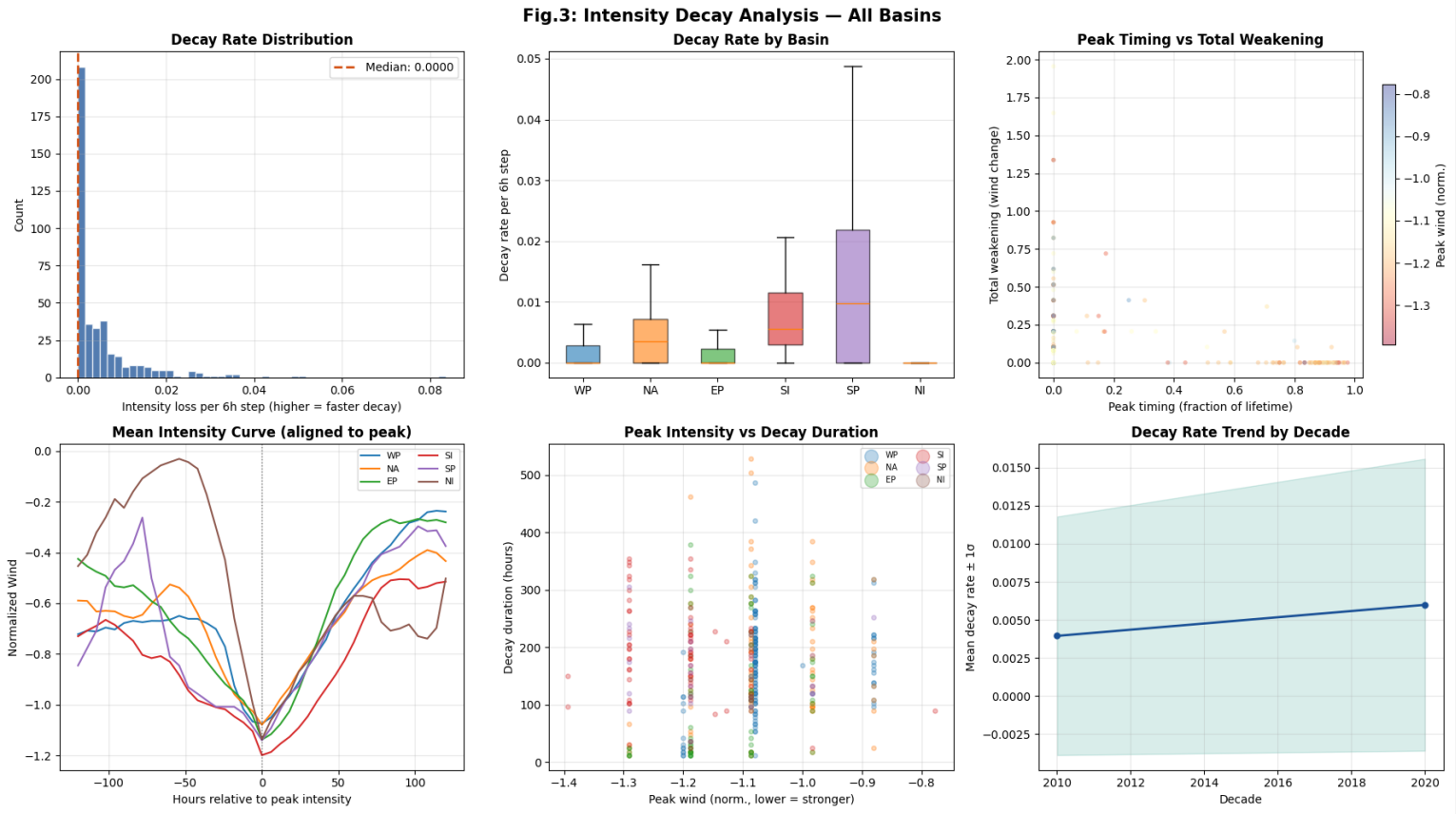
The "Decay Rate by Basin" plot shows that decay rate distribution varies widely by basin, implying the need for more nuanced model approaches.

Peak Intensity as a Poor Predictor:

The "Peak Intensity vs. Decay Duration" plot shows that the peak intensity alone is not a good predictor for decay duration, motivating the need to look into more features.

Extreme Decay Scenarios:

The "Decay Rate Distribution" plot shows that most storms weaken very gradually post-peak, but the tail shows there are extreme scenarios with sudden intensity drops.



Analysis

Spatial images overview

Landfall Images:

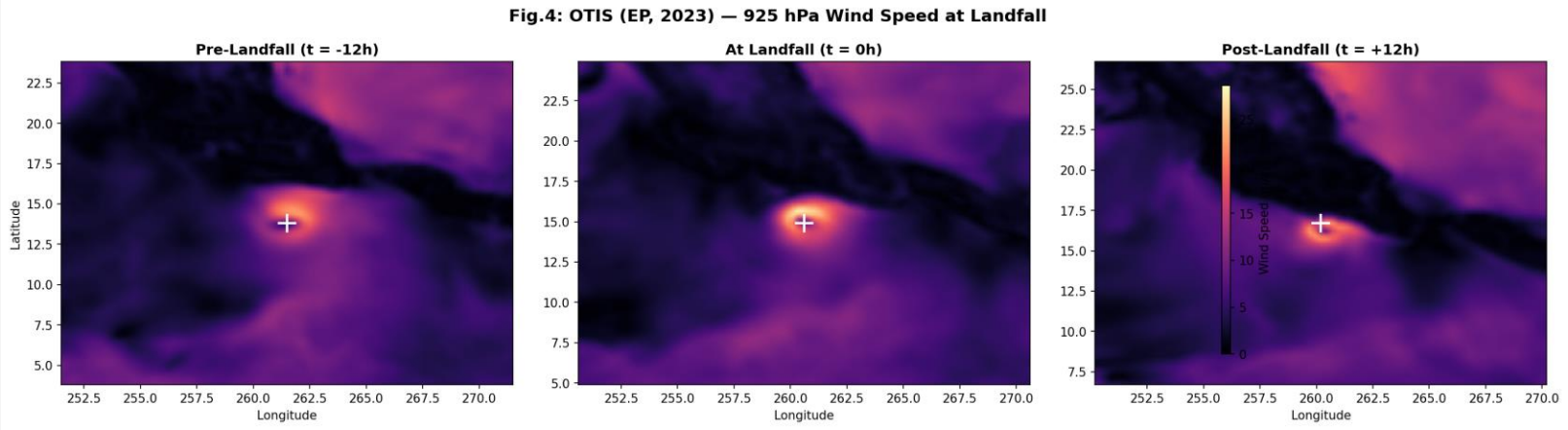
The wind field structure progressively collapses as the storm moves onshore.

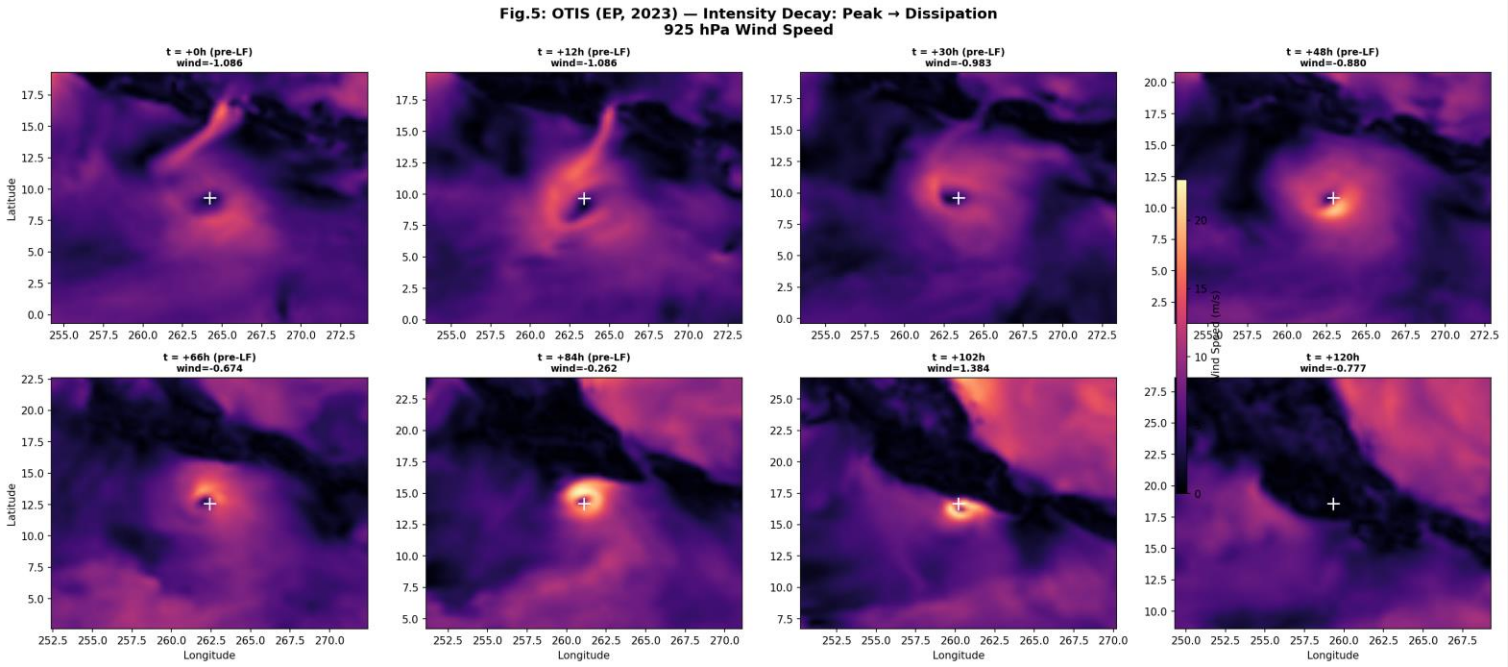
Decay Images:

Through these images, we can observe the storm core gradually weakening.

Takeaway:

These visualizations confirm that 925 hPa spatial wind data captures landfall dynamics that tabular
data alone cannot, motivating its use as a key model input.





Given our data observations, we need a physics-based model that can utilize spatial and tabular data.

There are many factors to consider, so feature selection strategies are imperative for understanding what will benefit the prediction task the most.

Model Architecture

Our U-FNO solution

Our U-FNO[1] architecture allows us to execute dual-domain feature extraction (spatial + spectral).

U-Net (Spatial): Captures **local, non-linear interactions** in the cyclone's inner core (e.g., eyewall formation, rainbands).

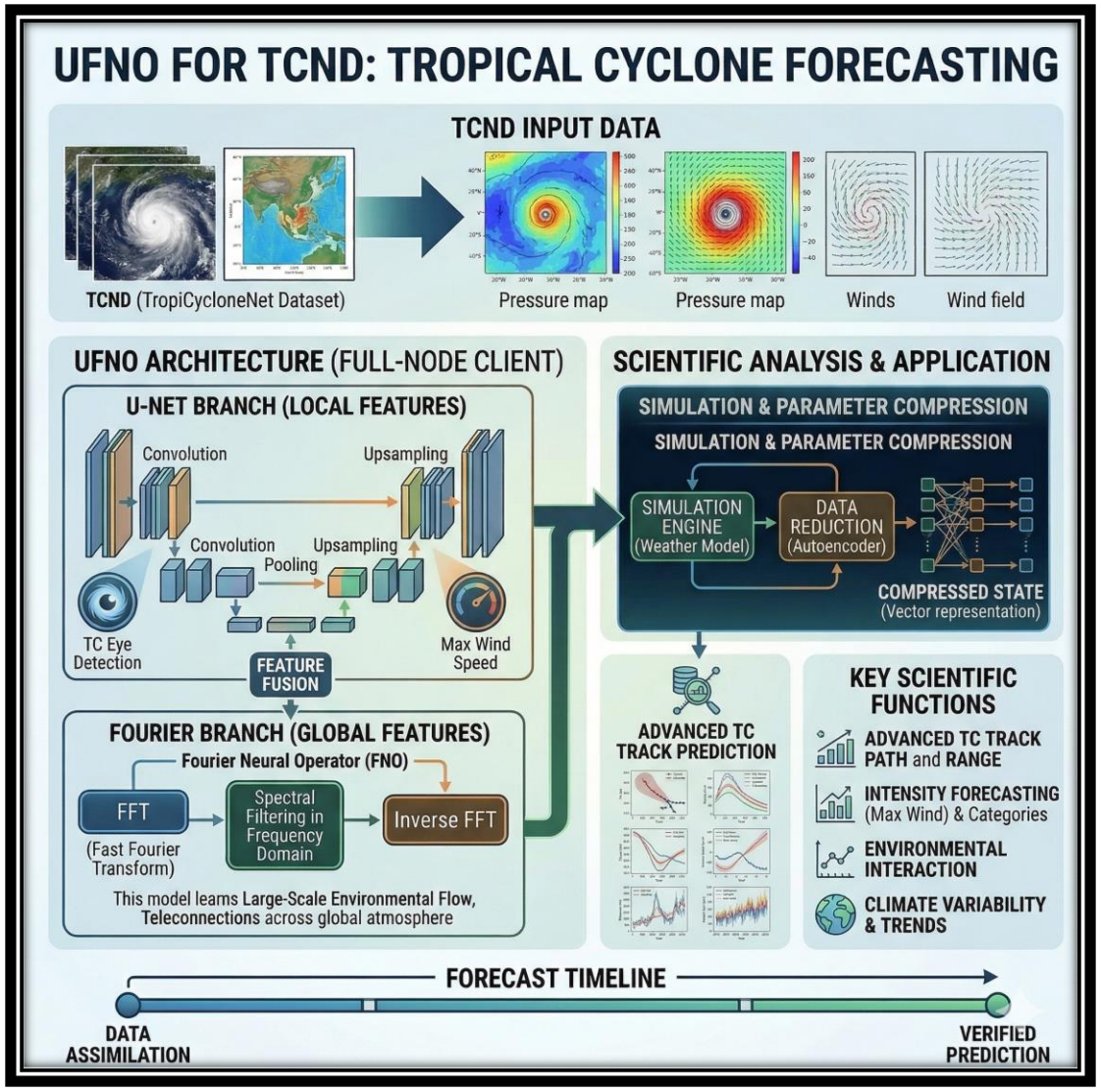
FNO (Spectral): Captures **long-range global dependencies**. By operating in the frequency domain

Fourier layer acts as an **integral kernel solver**
$G(u) = F^{-1}(R_\varphi \cdot F(u))$

Resolution Invariance (Zero-Shot Super-Resolution)

Computational Efficiency for "Quicker Prediction"

Multi-Scale Feature Fusion via Skip Connections



1. Wen et al. (2022)

Feature Ablation

Selecting the best features for prediction tasks

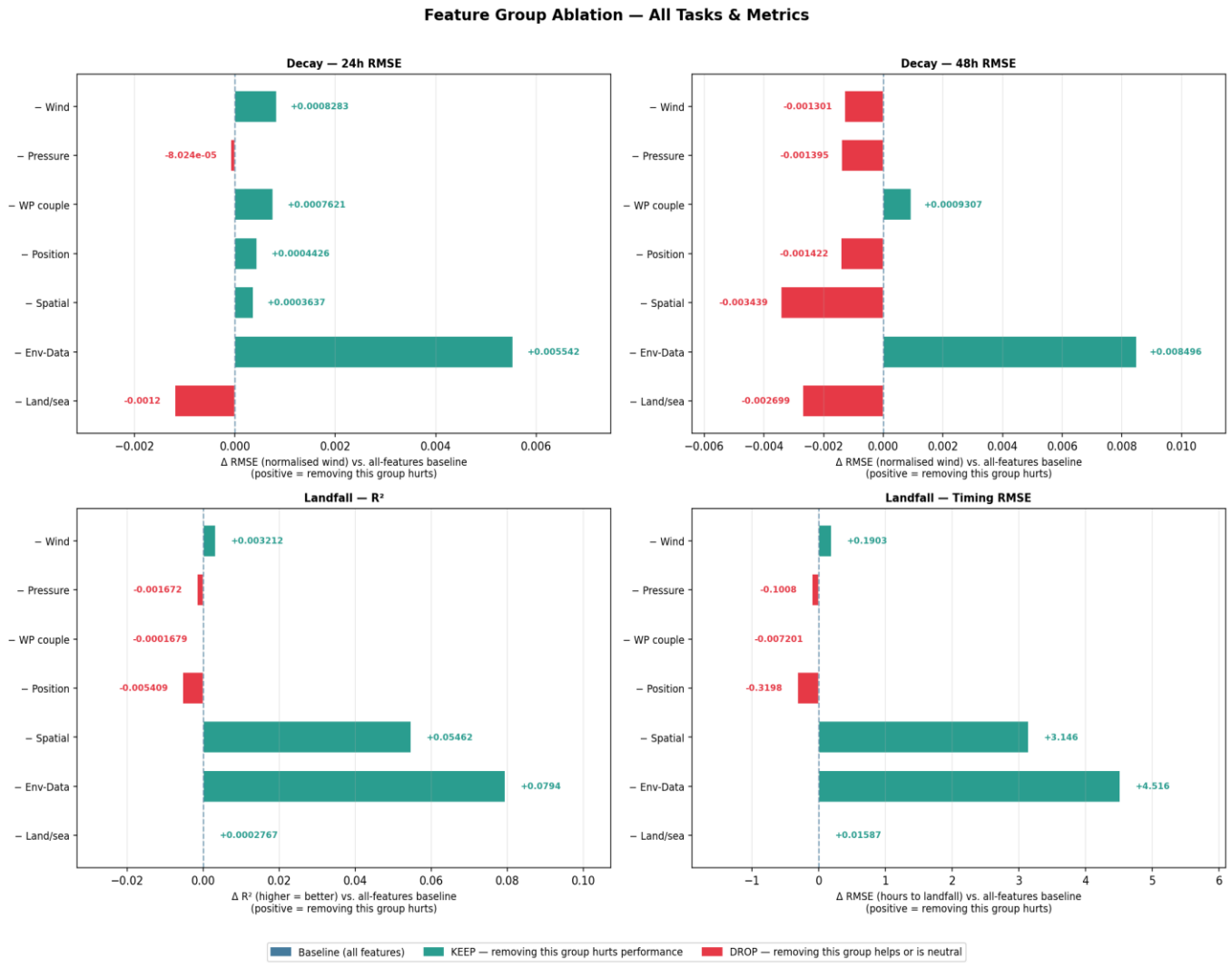
Feature ablation allowed us to determine the importance of each component to our prediction task.

Task: Using XGBoost, we remove one feature group at a time, and measure the change in RMSE

Goal: Feature ablation tells us where we can **simplify the architecture** without losing accuracy.

Results:

Rank	Intensity Decay (24h) (RMSE +)	Intensity Decay (48h) (RMSE +)	Landfall Prediction (R^2 +)	Landfall Timing (RMSE +)
1st Best	Env-Data (+0.005542)	Env-Data (+0.008496)	Env-Data (+0.0794)	Env-Data (+4.516)
2nd Best	Wind (+0.0008283)	WP Coupling (+0.0009307)	Spatial (numerical) (+0.05462)	Spatial (numerical) (+3.146)
3rd Best	WP coupling (+0.0007621)	N/A (rest were negative)	Wind (+0.003212)	Wind (+0.1903)



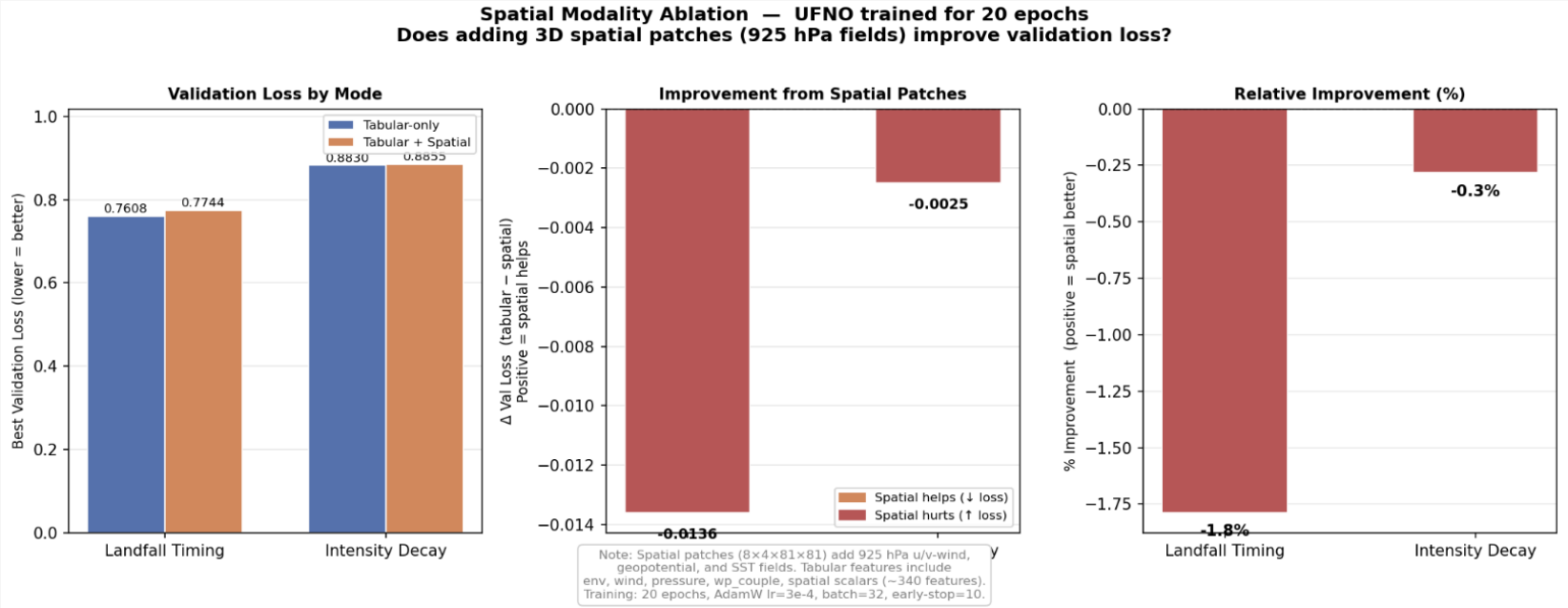
Feature ablation with our spatial images also revealed that adding the 3D spatial patches did not improve validation loss.

Issue: XGBoost cannot process spatial images.

Goal: Devise an alternative method for testing feature ablation on the spatial image data in TCND.

Task:

1. We train the CycloneUFNO once using tabular features only, and once with tabular features augmented by 925 hPa spatial patch tensors.
2. We then compare the best validation loss across both modes.



Findings: Surprisingly, the 3D spatial patches did not seem to improve model prediction performance in both tasks.

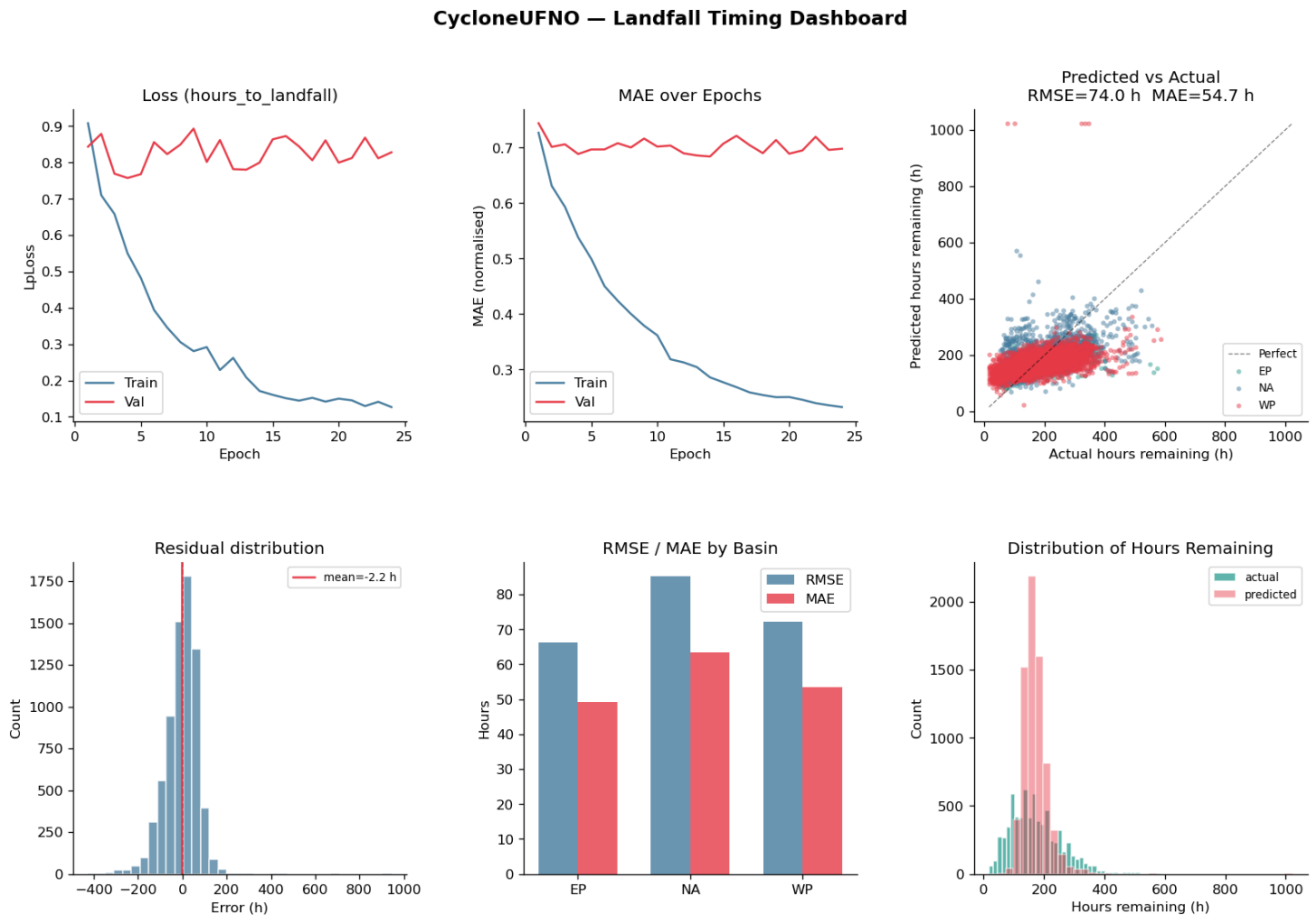
- Landfall timing showed a marginally lower performance/higher validation loss value, which **contrasts with our findings that spatial *numerical* features were important.**
- Intensity decay showed little to no improvement.

Model Results

Outcomes from predicting landfall impact and intensity decay

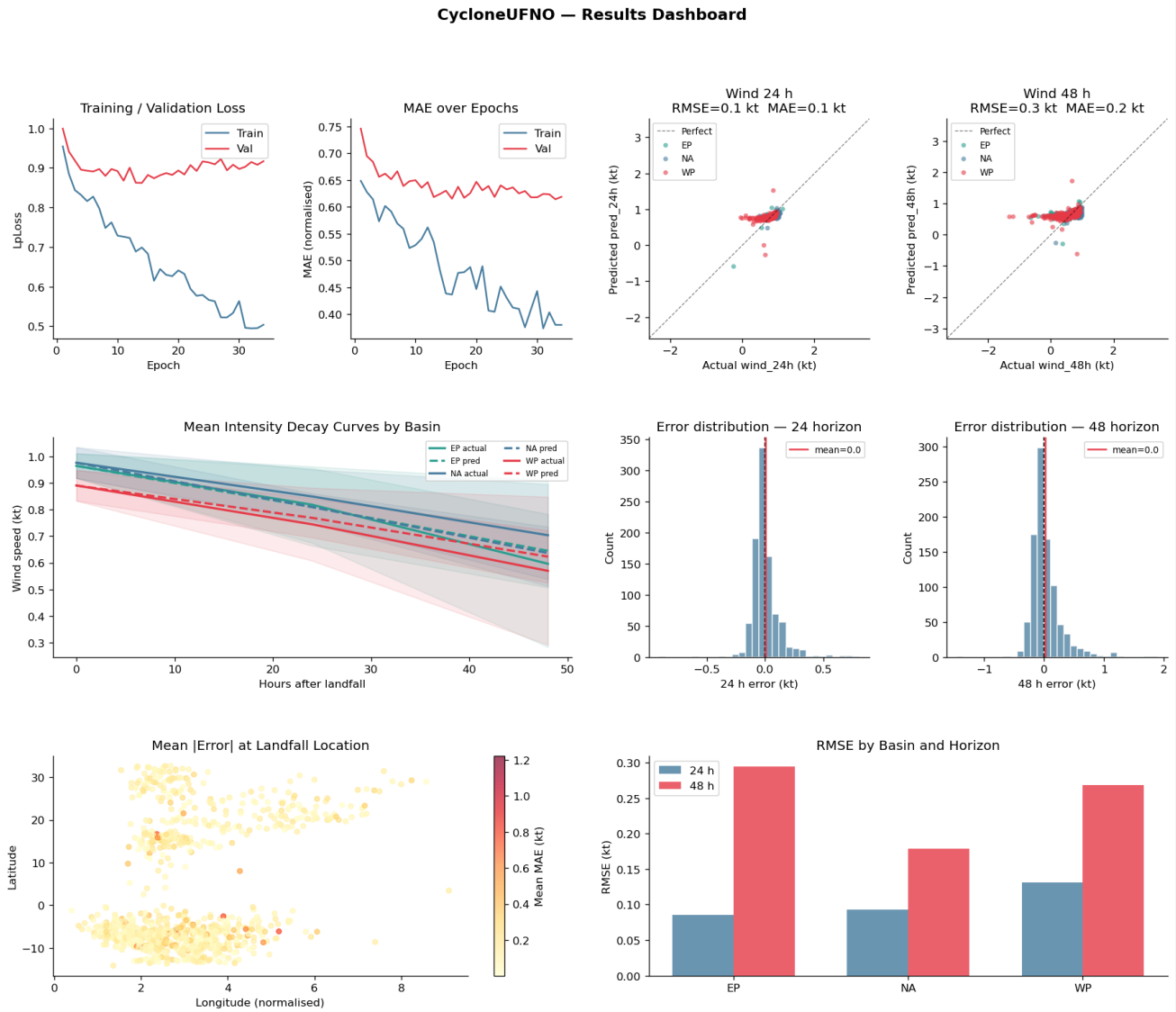
Our model for the landfall timing prediction task tends towards the mean instead of capturing the variability observed in real-world data.

- **Significant findings include:**
  - RMSE of 74 hours, **failing to close the prediction gap** we aimed to address.
  - **Significant overfitting**, train loss converges while validation loss plateaus.
  - Model collapses toward the mean
    - For actual values greater than 200 hours, the model tends to under-predict, **not performing well for storms far from landfall.**
    - The predicted distribution is far narrower than the actual, suggesting that the **model rarely predicts extreme values.**



Our model for intensity decay prediction performed better.

- **Significant findings include:**
  - Like the landfall timing prediction model, there is **overfitting, but it is less severe.** Despite this, **performance is rather good.**
    - RMSE = 0.1 kt at 24 hours and 0.3 kt at 48 hours. Error distribution is also tightly centered on 0.
  - The RMSE by basin and horizon shows that **longer horizons are harder to predict.**
  - Decay curves by basin deviate slightly from the actual curve, but **capture its general shape.**
  - The MAE by location plot shows no geographic pattern, confirming **spatial location is not a key predictor.**



Overfitting persists across all model sizes.

Intensity Decay

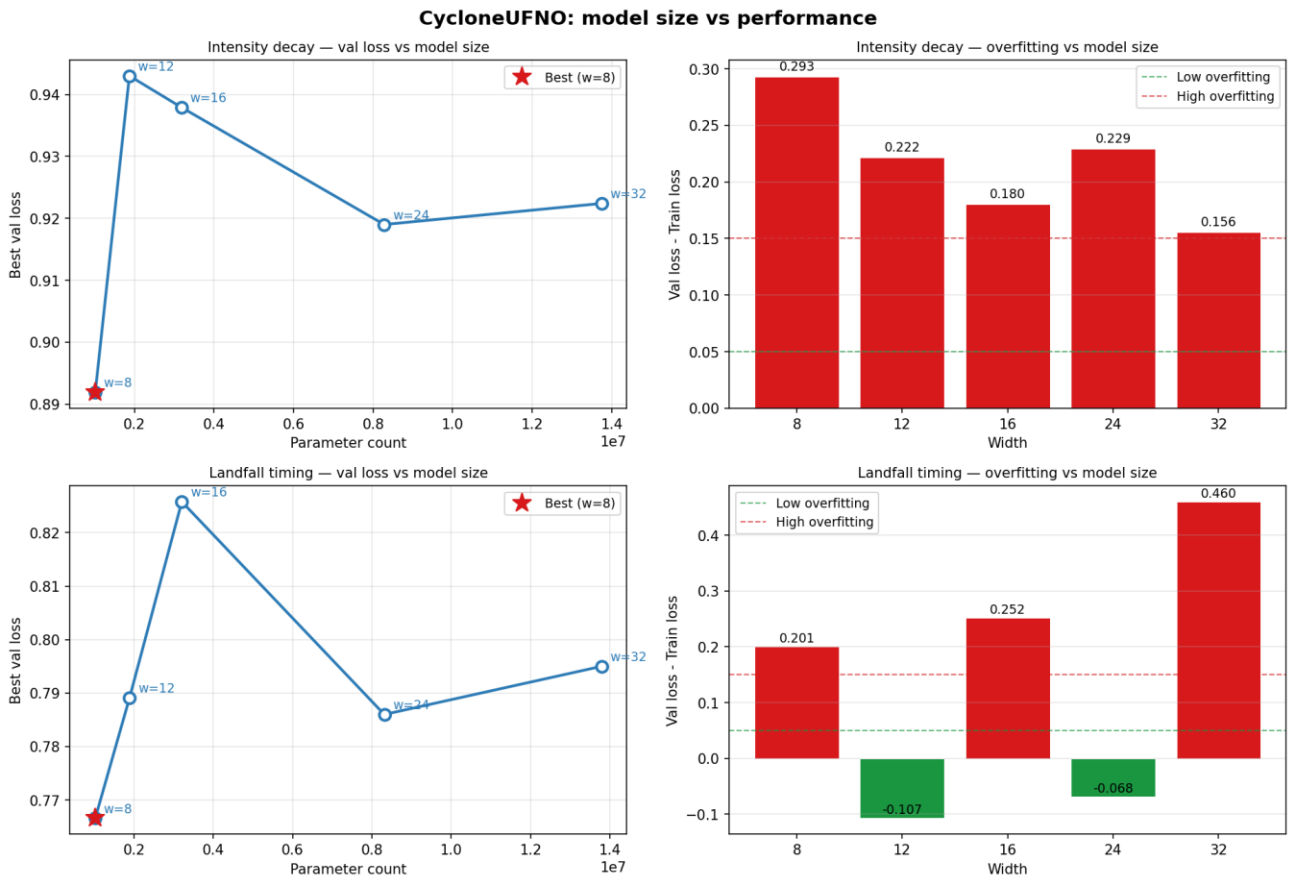
- Overfitting across all model sizes.

Landfall Timing

- Overfitting and underfitting, showing that the model can't fit the training data consistently.

Conclusion

- Model capacity is not the constraint.
- Therefore, this points to other factors as the root cause of the poor performance.

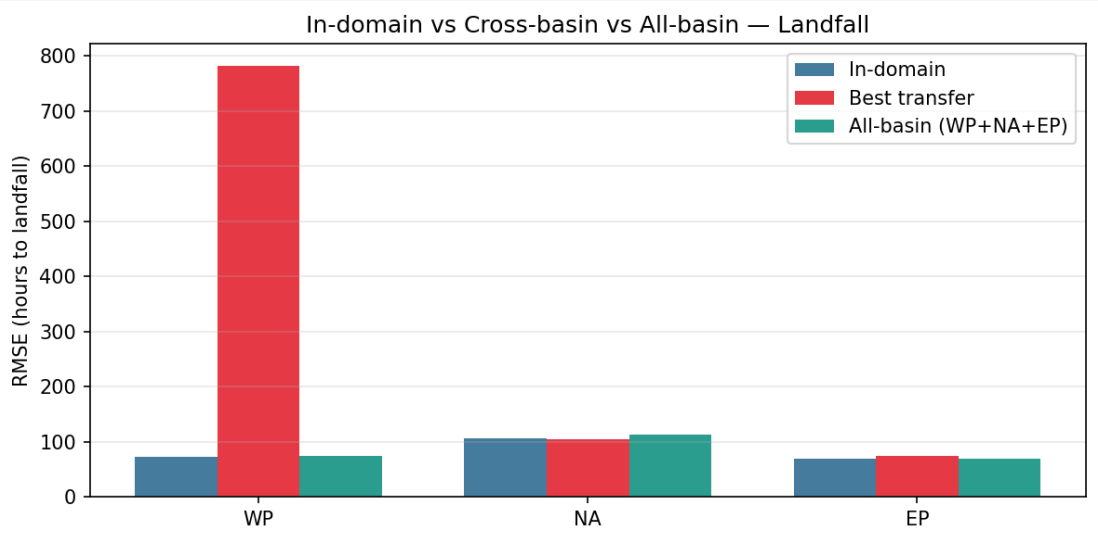


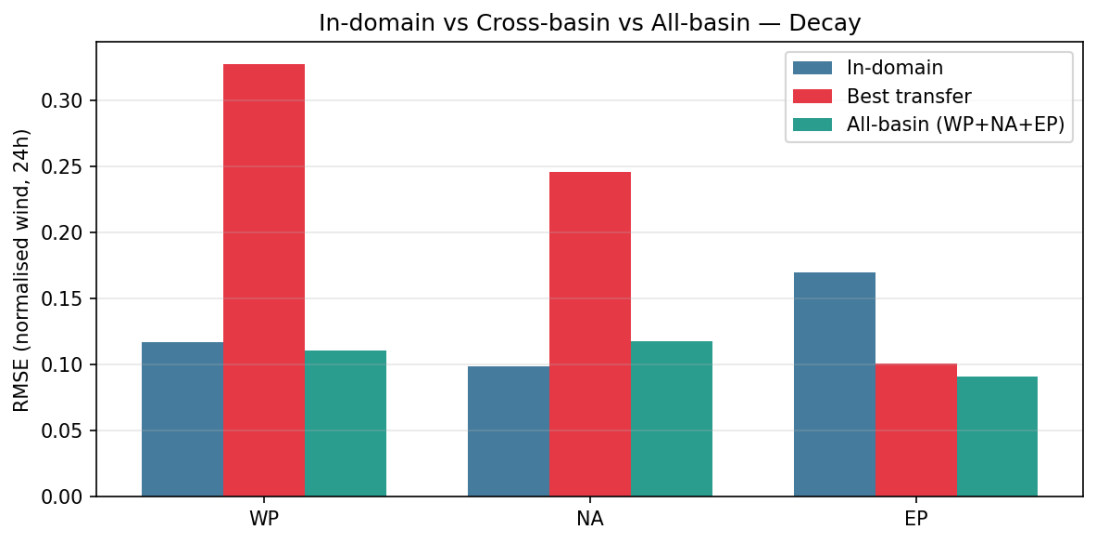
What could explain the pitfalls in the performance of our predictive models?

Cross-Basin Generalization

Understanding model pitfalls

Running cross-basin generalization allowed us to diagnose one potential issue contributing to our model's mediocre performance.





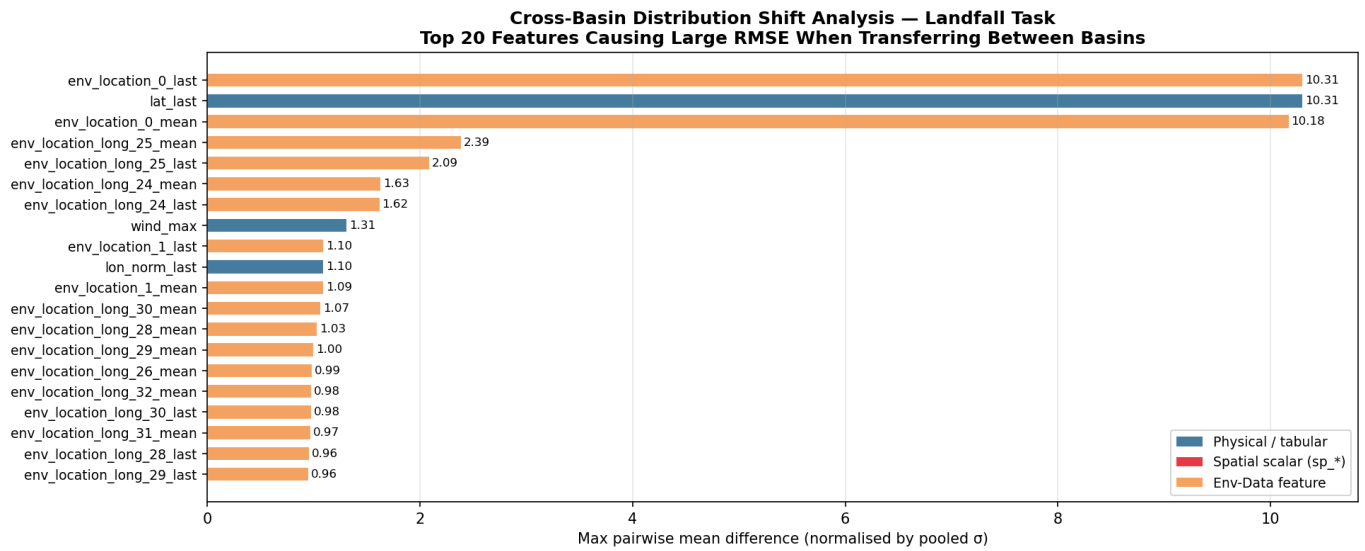
- **Cross-basin transfer for the WP basin in landfall timing fails catastrophically,** which may suggest that the WP basin has specific cyclone dynamics that cannot be approximated by other basin data.
- **Cross-basin transfer for intensity decay is better, but still poor**. With the exception of the EP basin, the transfer condition yields the highest RMSE.
- For both tasks, **almost all basins show that training on the in-domain basin v.s all basins yields similar results,** which opens some interesting questions.
  - One idea is that the all-basin model captures the nuances across basins.
  - Another idea is that both models are hitting a failure mode that limits their performance (data scarcity for the in-domain model, distribution shift for the all-basin model).

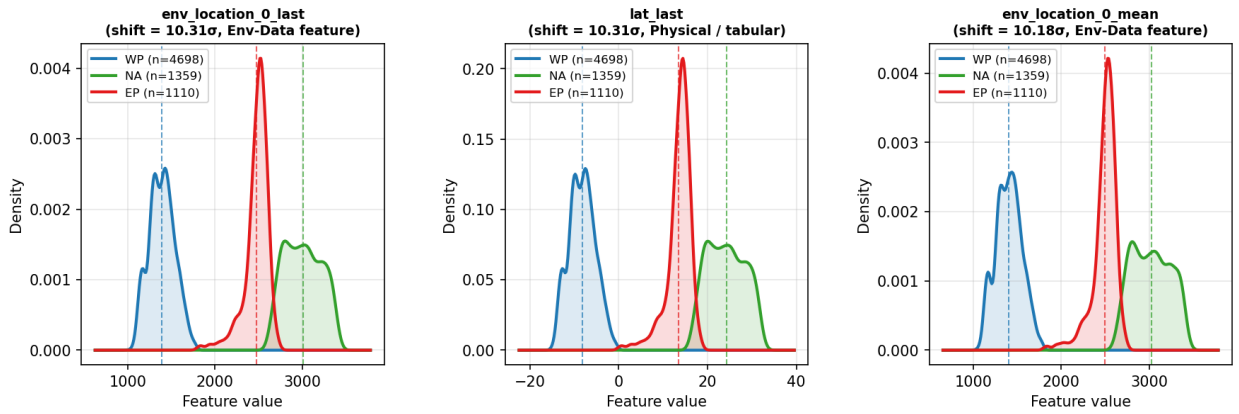
Distribution shifts regarding the locations of tropical storms in each basin were the primary culprit.

- The **Coriolis Force**, which controls how storms rotate and intensify, depends on latitude.
  - Since the latitude distribution of each basin varies widely, storms in each basin will behave very differently.
- **These findings reinforce what we found in the cross-basin generalization tests.**
  - A model trained on one basin is learning physics from a different latitude regime, making cross-basin generalisation fundamentally difficult.

$$f = 2\Omega \sin(\theta)$$

θ = latitude
f = the Coriolis parameter
Ω = Earth's rotation rate





These findings highlight the need for location-specific models to overcome the data distribution shift.

Discussion & Conclusion

Our final takeaways

We encountered several limitations with the TCND and the Modeling Methodology.

Data

- No ground-truth landfall timestamps, so landfall crossing needed to be approximated.

Methodology

- For storms that never made landfall, the "time to landfall" field was set to whenever the storm ended.
- The landfall timing task may be better suited to a time-sequencing model such as an LSTM.
- The distribution shift findings show that our original all-basin approach may mask the nuances across basins, thus motivating future work in training location-specific models.

Our project introduced us to the various complexities that must be considered when devising ML systems for tackling climate change.

- **Our choice of the U-FNO model was problem and data-driven.**
- **Feature ablation for tabular/spatial numerical/environmental data and 3D spatial images showed nuanced findings.**
  - For example, spatial numerical data was marked as an important feature, but 3D spatial images were not.
- **Our intensity decay model performed better than our landfall timing model,** but both failed to predict extreme cases (storms further away from landfall and scenarios with longer horizons).
  - Intensity decay labels are directly observed measurements, whereas landfall timing labels are approximated due to the lack of ground-truth data
- The cross-basin generalization tests highlighted how **distribution shifts in latitude and location of the storm** contribute to limitations in the model performance.

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Thank you.

References

Shi, Y., Hu, R., Wu, N., Zhang, H., Liu, X., Zeng, Z., ... & Shi, X. (2025). Comparison of AI and NWP models in operational severe weather forecasting: A study on tropical cyclone predictions. Journal of Geophysical Research: Machine Learning and Computation, 2(2), e2024JH000481.

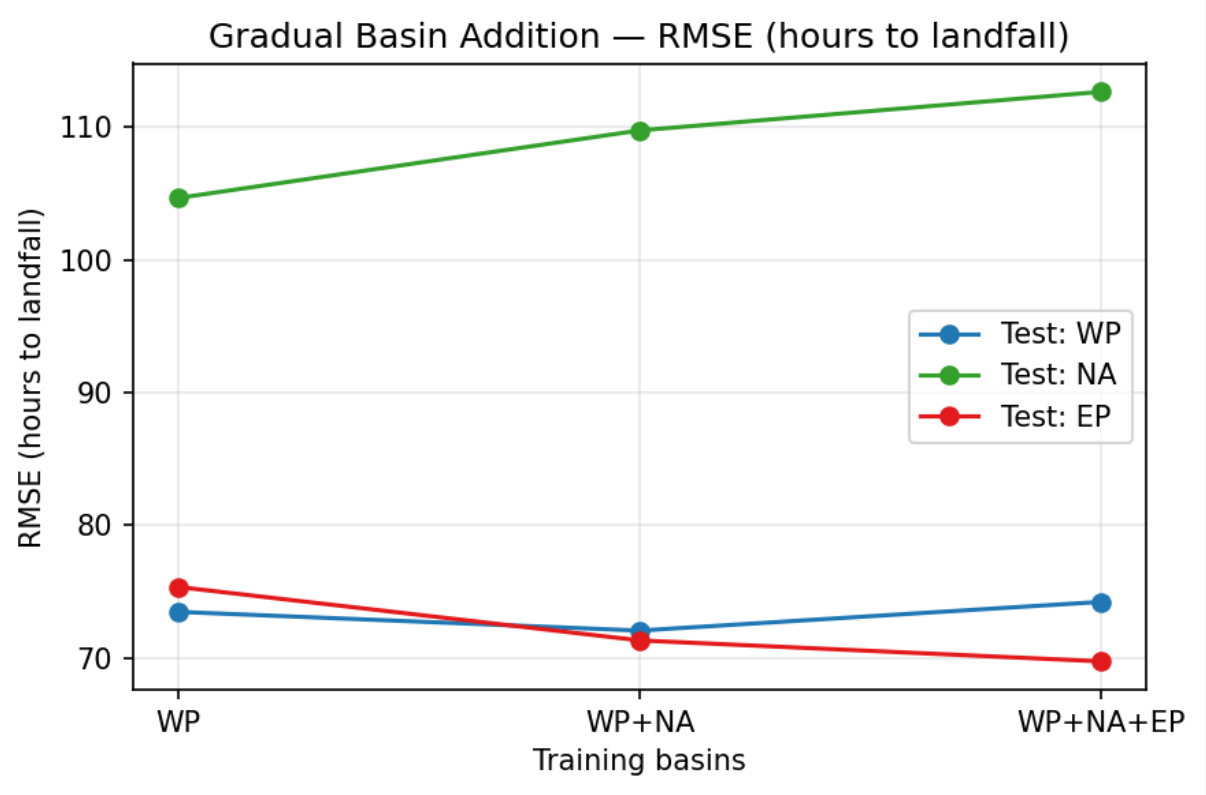
Wen, G., Li, Z., Azizzadenesheli, K., Anandkumar, A., & Benson, S. M. (2022). U-FNO—An enhanced Fourier neural operator-based deep-learning model for multiphase flow. Advances in Water Resources, 163, 104180.

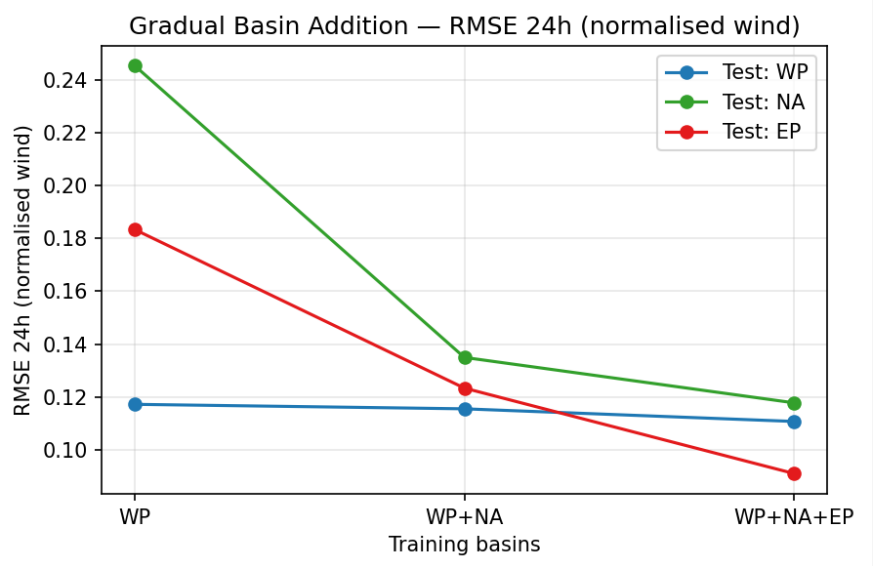
https://www.climate.gov/maps-data/climate-data-primer/predicting-climate/climate-models

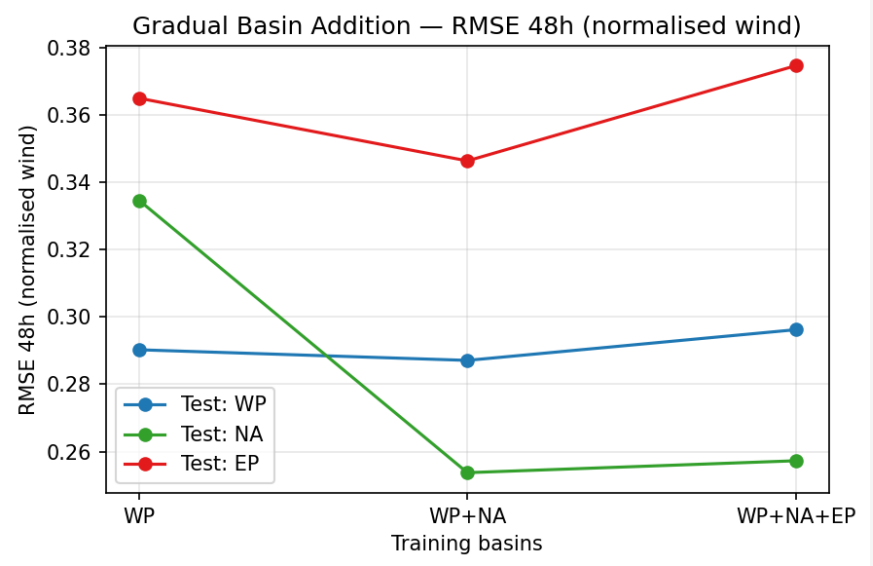
Appendix

Additional Slides

Cross-Basin Gradual Training Results





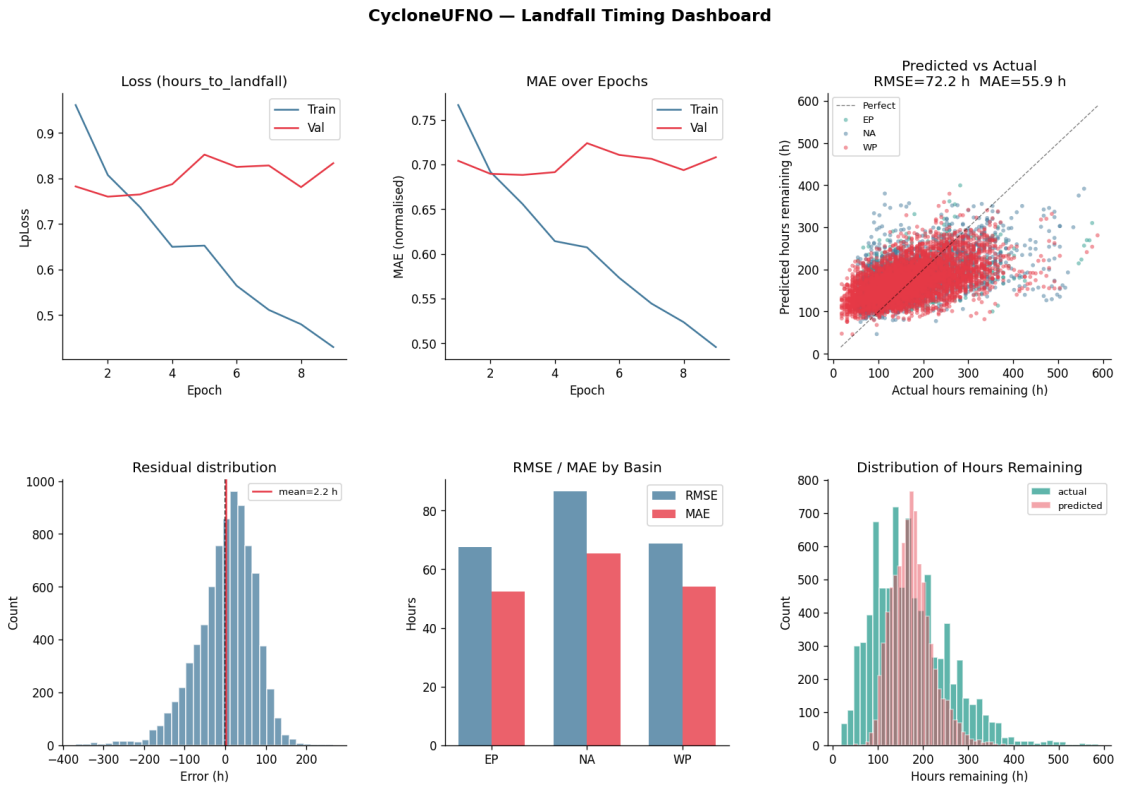


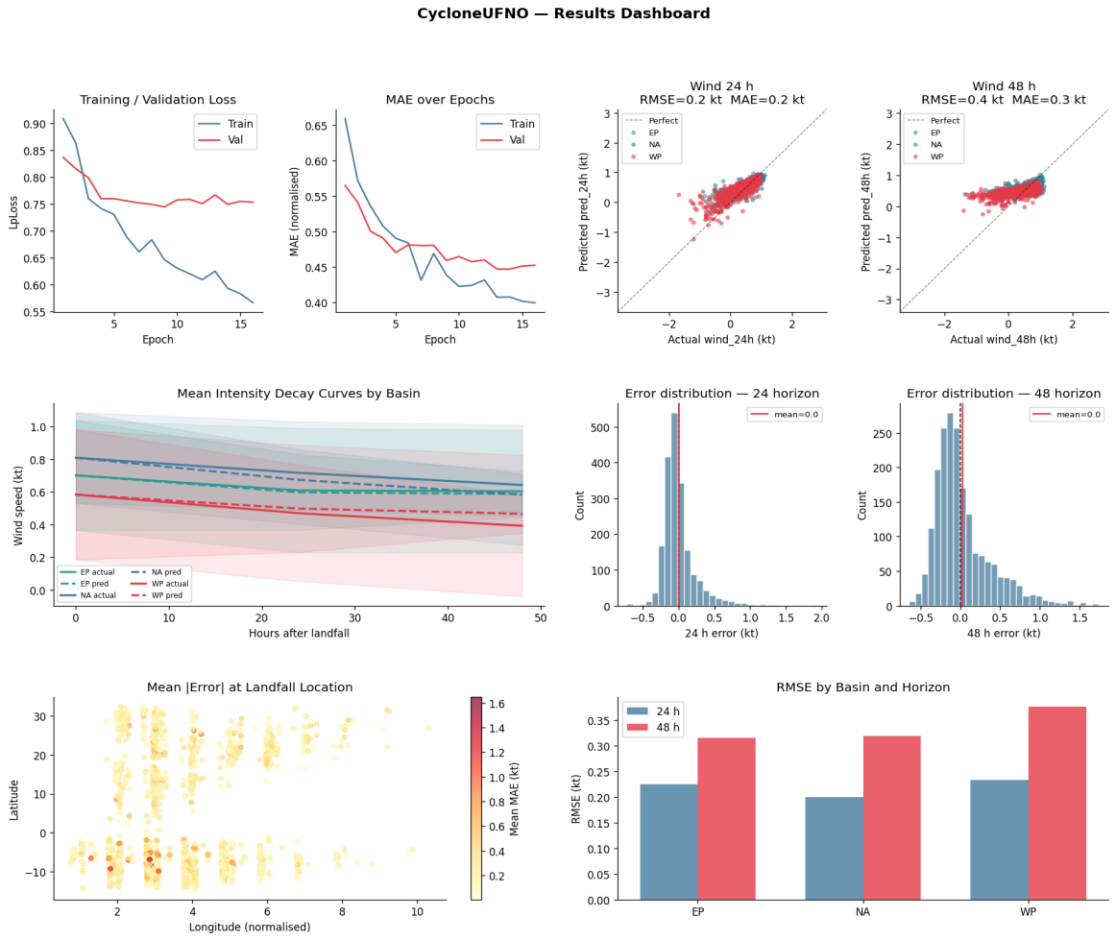
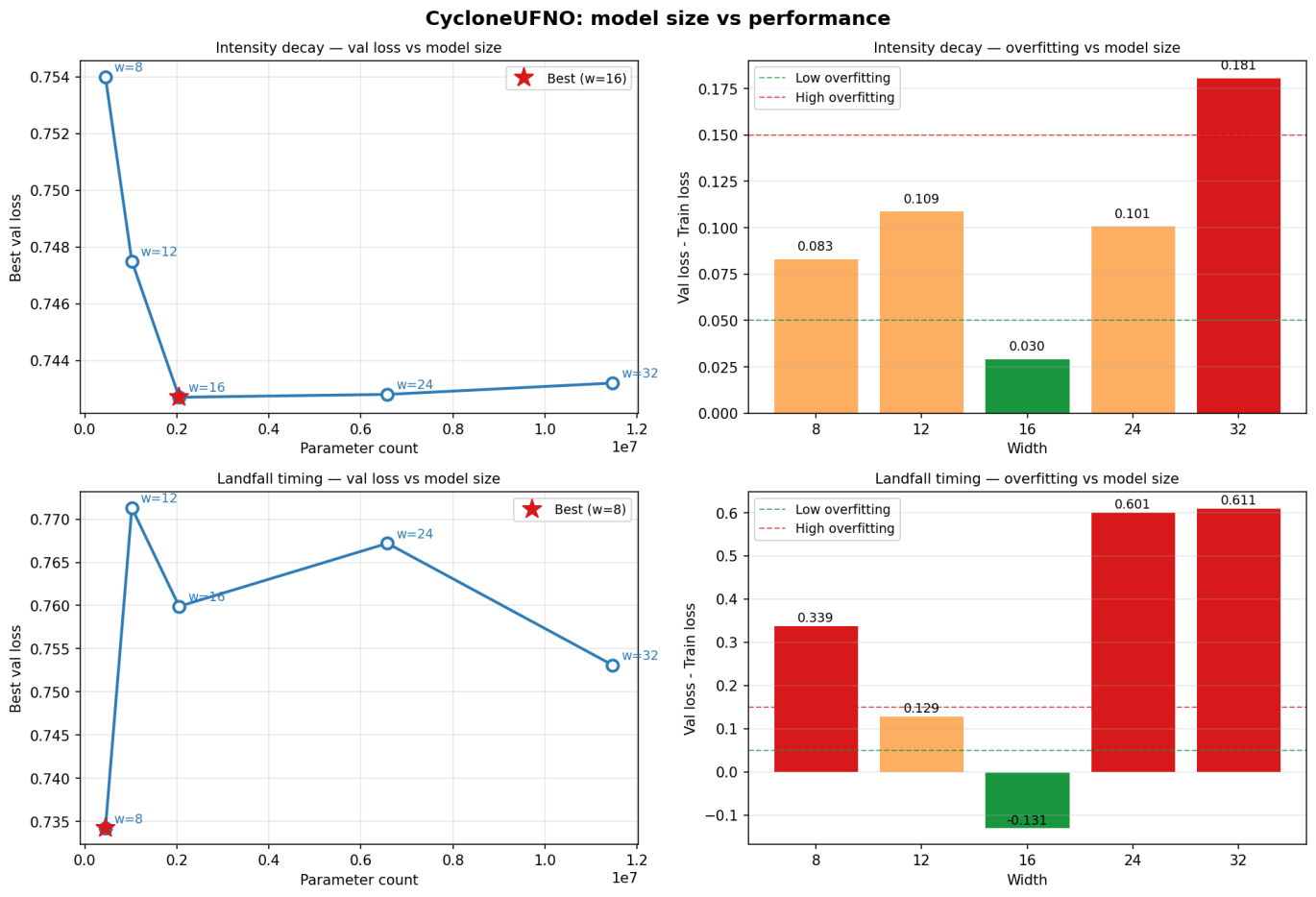
Rotational Component of Coriolis Force

$$\mathbf{\Omega} = \omega \begin{pmatrix} 0 \\ \cos\varphi \\ \sin\varphi \end{pmatrix}, \mathbf{v} = \begin{pmatrix} v_\mathrm{e} \\ v_\mathrm{n} \\ v_\mathrm{u} \end{pmatrix},$$

$$\mathbf{a}_\mathrm{C} = -2\mathbf{\Omega} \times \mathbf{v} = 2\,\omega \begin{pmatrix} v_\mathrm{n} \sin\varphi - v_\mathrm{u} \cos\varphi \\ -v_\mathrm{e} \sin\varphi \\ v_\mathrm{e} \cos\varphi \end{pmatrix}.$$

Final Experiment: Cropping Images to 20x20, Collapsing Env Features

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